BUSINESS REQUIREMENTS DOCUMENT

Customer Churn Prediction and Retention Targeting

BRD-01 | Intermediate Data Science Project

Document ID	BRD-01
Version / Status	1.0 | Draft for approval
Date	8 November 2026
Business sponsor	Head of Customer Retention, ConnectPlus Telecom (fictional)
Prepared by	Data Science Team
Dataset	IBM Telco Customer Churn (Kaggle), 7,043 customers
Estimated effort	3 weeks (part-time)

1. Executive Summary

ConnectPlus Telecom (fictional) loses roughly one in four subscribers and currently sends retention offers without prioritisation. This project will build a machine learning solution that predicts which customers are likely to churn, explains why, and recommends whom to contact so the retention budget is spent where it returns the most value.

2. Business Problem and Opportunity

- Retention campaigns go to broad lists, so offers reach customers who would have stayed anyway.
- There is no visibility into churn drivers such as contract type, tenure, services and billing.
- Contacting the highest-risk customers first improves campaign ROI and reduces revenue loss.

3. Objectives and Success Criteria

Objective	Measure	Target
Predict churn accurately	ROC-AUC on hold-out set	≥ 0.83
Prioritise efficiently	Top-decile lift vs random targeting	≥ 2.5×
Catch at-risk customers	Churn-class recall at chosen threshold	≥ 0.70
Limit wasted offers	Churn-class precision at same threshold	≥ 0.50
Prove business value	Simulated campaign net benefit vs random targeting	Positive and quantified
Explain decisions	Top 3 churn drivers per scored customer	100% of customers

4. Scope

In scope

- Data cleaning, EDA and feature engineering
- Baseline logistic regression plus Random Forest and gradient boosting (XGBoost or LightGBM)
- Class-imbalance handling and cost-based threshold selection
- SHAP explainability and subgroup fairness check
- Streamlit dashboard for retention teams

Out of scope: real-time system integration, uplift (causal) modelling, running live A/B tests.

5. Stakeholders

Role	Interest
Head of Customer Retention (Sponsor)	Campaign ROI and prioritised call lists
Marketing Manager	Offer design and segment insights
Finance Analyst	Cost-benefit assumptions
Compliance Officer	Fairness and data handling
Data Science Lead (Owner)	Model quality and delivery

6. Business Requirements

Priority key (MoSCoW): M = Must have, S = Should have, C = Could have.

ID	Requirement	Priority
BR-01	Score every customer with a churn probability	M
BR-02	Assign each customer to a High, Medium or Low risk tier	M
BR-03	Provide per-customer explanation of top risk drivers	M
BR-04	Select the decision threshold using a cost-benefit model (offer cost, offer success rate, customer value)	M
BR-05	Dashboard to filter customers by risk and export contact lists	S
BR-06	Compare model performance across gender and senior-citizen groups	C

7. Data Requirements

- Source: IBM Telco Customer Churn dataset (Kaggle), 7,043 customers, 21 columns.
- Target: Churn (Yes/No), about 26.5% positive.
- Feature groups: demographics, account details (tenure, contract, payment method, charges), subscribed services.
- Known issues: TotalCharges is stored as text with 11 blank values (new customers with zero tenure); class imbalance; many categorical fields.
- Splitting: stratified 80/20 hold-out, with 5-fold stratified cross-validation on the training portion.

Full dataset details, data dictionary and download instructions are in **Appendix A** at the end of this document.

8. Assumptions, Constraints and Risks

Assumptions (illustrative and configurable): a retention offer costs 20 USD per contacted customer, 30% of contacted churners accept, and customer value equals 12 months of MonthlyCharges. Constraint: the data is a single snapshot, so no temporal validation is possible.

Risk	Mitigation
Class imbalance skews metrics	Stratified splits, class weights, PR-AUC alongside ROC-AUC
Overfitting on a small dataset	Cross-validation, regularisation, simple baseline for comparison
Data leakage	Fit all preprocessing inside pipelines; audit features
Biased outcomes for subgroups	Report metrics by subgroup

9. Deliverables and Timeline

Deliverables: EDA and modelling notebooks, modular Python code, trained model file, SHAP report, cost-benefit analysis, Streamlit app, model card and README.

Week	Focus
1	Cleaning, EDA, baseline model
2	Feature engineering, model comparison, tuning, imbalance handling
3	Explainability, cost analysis, dashboard, documentation

10. Acceptance Criteria

- Targets in Section 3 are met on the untouched hold-out set.
- The project runs end to end from the README in a clean environment.
- SHAP explanations and the cost-benefit analysis are delivered.
- A leakage review is documented and the sponsor has reviewed the dashboard demo.

11. Approvals

Role	Name	Signature	Date
Business Sponsor			
Project Owner (Data Science Lead)			
Reviewer			

Appendix A: Dataset

This appendix describes the dataset to be used for BRD-01 so the project can be started immediately.

A.1 Source and access

Dataset name	Telco Customer Churn (IBM sample data, hosted on Kaggle)
Kaggle page	https://www.kaggle.com/datasets/blastchar/telco-customer-churn
Original IBM source	https://community.ibm.com/community/user/blogs/steven-macko/2019/07/11/telco-customer-churn-1113
Access	Free Kaggle account needed for download. Review the license shown on the Kaggle page before redistributing the data.
Granularity	One row per customer (single snapshot, no date column)

A.2 Files and structure

File	Contents
WA_Fn-UseC_-Telco-Customer-Churn.csv	7,043 rows x 21 columns (about 1 MB). Target column: Churn; about 26.5% of customers churned.

A.3 Data dictionary

Field	Type	Description
customerID	string	Unique customer identifier (drop before modelling)
gender	categorical	Male / Female
SeniorCitizen	binary 0/1	Whether the customer is a senior citizen
Partner, Dependents	Yes / No	Whether the customer has a partner / dependents
tenure	integer	Number of months the customer has stayed with the company
PhoneService	Yes / No	Whether the customer has phone service
MultipleLines	categorical	Yes / No / No phone service
InternetService	categorical	DSL / Fiber optic / No
OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies	categorical	Yes / No / No internet service
Contract	categorical	Month-to-month / One year / Two year
PaperlessBilling	Yes / No	Whether the customer uses paperless billing
PaymentMethod	categorical	Electronic check / Mailed check / Bank transfer (automatic) / Credit card (automatic)
MonthlyCharges	float	Amount charged to the customer each month
TotalCharges	text (convert to float)	Total amount charged to date; 11 blank values for customers with tenure 0
Churn (target)	Yes / No	Whether the customer left within the last month

A.4 Download and load

```
# Option 1: Kaggle CLI (needs a Kaggle API token in ~/.kaggle/kaggle.json)
kaggle datasets download -d blastchar/telco-customer-churn --unzip

# Option 2: kagglehub (Python)
import kagglehub
path = kagglehub.dataset_download("blastchar/telco-customer-churn")

# Load and fix the known data-type issue
import pandas as pd
df = pd.read_csv(path + "/WA_Fn-UseC_-Telco-Customer-Churn.csv")
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
df["Churn"] = (df["Churn"] == "Yes").astype(int)
```

A.5 Data notes and caveats

- TotalCharges contains blank strings for 11 customers with tenure 0. Convert to numeric and impute with 0 or drop, and document the choice.
- The dataset has no time dimension, so use stratified cross-validation rather than time-based splits.
- Features such as gender and SeniorCitizen are sensitive attributes: use them for fairness checks, and justify any use as model inputs.
- This is a sample dataset for a fictional company; business cost parameters in Section 8 are assumptions, not observed values.